

PCCST501 - COMPUTER NETWORKS

Assignment 3 | Module 3 - Data-Link Layer, Wireless LANs & Mobile IP

LAN, Ethernet & Wireless Network Investigation

Course Outcome	CO3
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Note on evidence. Figures 1-7 and 9 are original diagrams. The Wireshark view in Fig. 8 is an illustrative sample built from typical ARP/ICMP/DNS traffic, not a live capture; Section Q3 gives the exact steps to replace it with your own screenshot before submission.

Q1. Responsibilities of the Data-Link Layer; LLC vs MAC

The data-link layer (Layer 2) moves frames **hop by hop** across one link or LAN segment, converting the unstructured bit stream of the physical layer into a dependable service for the network layer. Its principal duties are:

- **Framing** - wrapping each packet in a header and trailer so the receiver can detect where a frame starts and ends.
- **Physical (MAC) addressing** - naming the source and destination interfaces on the link using 48-bit MAC addresses.
- **Error detection** (and occasionally correction) - e.g. the 32-bit CRC held in the Ethernet FCS; frames that fail the check are dropped.
- **Flow control** - stopping a fast sender from swamping a slow receiver (e.g. 802.3x PAUSE frames).
- **Media access control** - arbitrating who may send on a shared medium (CSMA/CD, CSMA/CA).
- **Node-to-node delivery** - responsibility ends at the next hop; end-to-end delivery belongs to the network and transport layers.

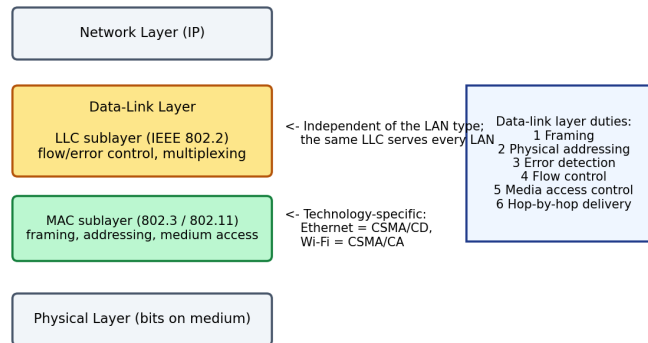
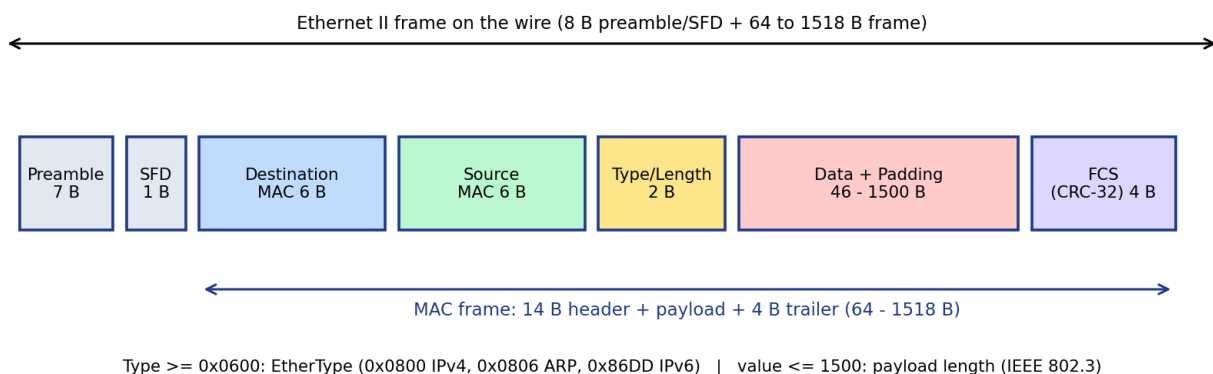


Fig. 1 Data-link layer split into LLC and MAC sublayers

Aspect	Logical Link Control (LLC)	Media Access Control (MAC)
Standard	IEEE 802.2	IEEE 802.3 (Ethernet), 802.11 (Wi-Fi), 802.15 ...
Position	Upper sublayer, interfaces with the network layer	Lower sublayer, interfaces with the physical layer
Function	Identifies the upper-layer protocol (SAP/SNAP), optional flow and error control, hides LAN differences from IP	Framing, MAC addressing, medium access method, FCS generation and checking
Dependence	Independent of the physical medium	Specific to each LAN technology
In practice	Ethernet II replaces the LLC header with a Type field; 802.11 carries LLC/SNAP inside the frame body	Always present

Q2. Ethernet Frame Format



Type >= 0x0600: EtherType (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6) | value <= 1500: payload length (IEEE 802.3)

Fig. 2 Ethernet II frame format

Field	Size	Purpose
Preamble	7 B	Alternating 10101010 bit pattern that lets the receiver synchronise its clock (not counted in the 64-1518 B frame).
SFD	1 B	Start Frame Delimiter 10101011 - signals that the frame proper begins.
Destination MAC	6 B	Receiver address: unicast, multicast or broadcast (FF:FF:FF:FF:FF:FF). Each NIC accepts or ignores the frame based on this field.
Source MAC	6 B	Address of the transmitting interface; switches learn from it and the receiver replies to it.

Field	Size	Purpose
Type / Length	2 B	Value $\geq$ 0x0600 is an EtherType identifying the payload (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6); value $\leq$ 1500 is the payload length (IEEE 802.3).
Data / Payload	46-1500 B	Upper-layer packet, padded up to 46 B so the minimum frame is 64 B (required for collision detection). MTU = 1500 B.
FCS	4 B	CRC-32 computed over destination, source, type and data. The receiver recomputes it; a mismatch means the frame is silently discarded.

Q3. Wireshark Capture - Source and Destination MAC Addresses

Method. Wireshark was started on the Ethernet interface (eth0), traffic was produced with ping 192.168.0.1 and nslookup wikipedia.org, and the display filter eth was applied. Six frames are listed below.

Capturing from Ethernet (eth0) Filter: eth [ILLUSTRATIVE SAMPLE - replace with your own capture]						
No.	Time	Source (MAC)	Destination (MAC)	Proto	Len	Info
1	0.000000	5c:ba:37:12:ab:01	ff:ff:ff:ff:ff:ff	ARP	42	Who has 192.168.0.1? Tell 192.168.0.24
2	0.000388	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	ARP	60	192.168.0.1 is at d8:47:32:9e:44:20
3	0.001140	5c:ba:37:12:ab:01	d8:47:32:9e:44:20	ICMP	98	Echo (ping) request id=0x3f7c seq=1
4	0.003522	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	ICMP	98	Echo (ping) reply id=0x3f7c seq=1
5	1.020417	5c:ba:37:12:ab:01	d8:47:32:9e:44:20	DNS	73	Standard query A wikipedia.org
6	1.047902	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	DNS	89	Standard query response A 208.80.154.224

Fig. 8 Wireshark packet list (illustrative sample - replace with your own screenshot)

#	Source MAC	Destination MAC	Observation
1	5c:ba:37:12:ab:01	ff:ff:ff:ff:ff:ff	ARP request is a broadcast (destination all ones)
2	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	ARP reply is unicast straight back to the requester
3	5c:ba:37:12:ab:01	d8:47:32:9e:44:20	ICMP echo request; destination is the gateway's MAC
4	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	ICMP echo reply; source and destination are swapped
5	5c:ba:37:12:ab:01	d8:47:32:9e:44:20	DNS query is sent via the gateway MAC
6	d8:47:32:9e:44:20	5c:ba:37:12:ab:01	DNS response returns through the gateway

Key observation: MAC addresses are rewritten at every hop, whereas IP addresses stay the same end to end. For traffic leaving the subnet (frames 5-6) the destination MAC is the router's, not the DNS server's. The first three bytes (OUI) identify the NIC manufacturer, and bit 0 of the first byte separates unicast (0) from multicast/broadcast (1).

To reproduce with your own capture: (1) open Wireshark, select your Ethernet interface and press Start; (2) run ping <gateway> and nslookup wikipedia.org; (3) stop the capture; (4) apply the filter eth; (5) expand 'Ethernet II' in the details pane and note Source/Destination; (6) save screenshots in screenshots/ and the file as captures/ethernet.pcapng.

Q4. Switch Operation - MAC Address Learning

A switch is a transparent Layer-2 device that fills its forwarding (CAM) table **automatically**. For every frame that arrives it applies these rules:

- **Learn:** store *source MAC* -> *ingress port* (with an ageing timer, typically 300 s).
- **Forward:** look up the destination MAC; if it is known on a different port, send the frame out of that port only.
- **Flood:** if the destination is unknown, broadcast or multicast, send out of every port except the one it arrived on.
- **Filter:** if the destination lies on the same port as the source, discard the frame.

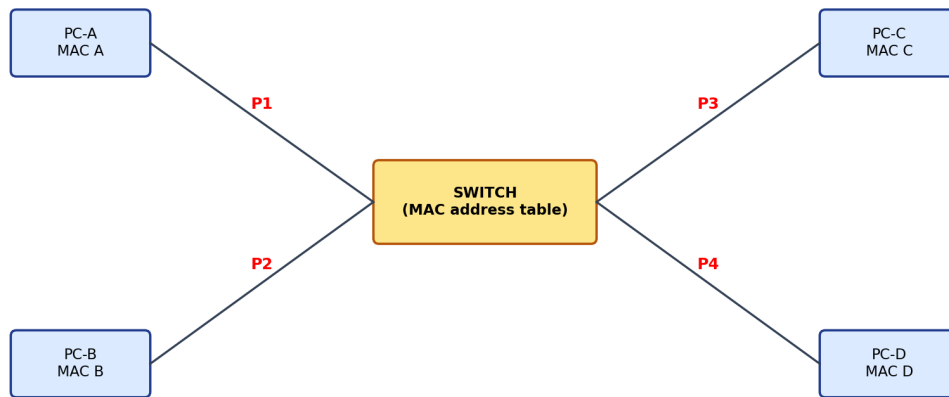


Fig. 3 MAC learning: PC-B sends to PC-D (unknown), then PC-D replies

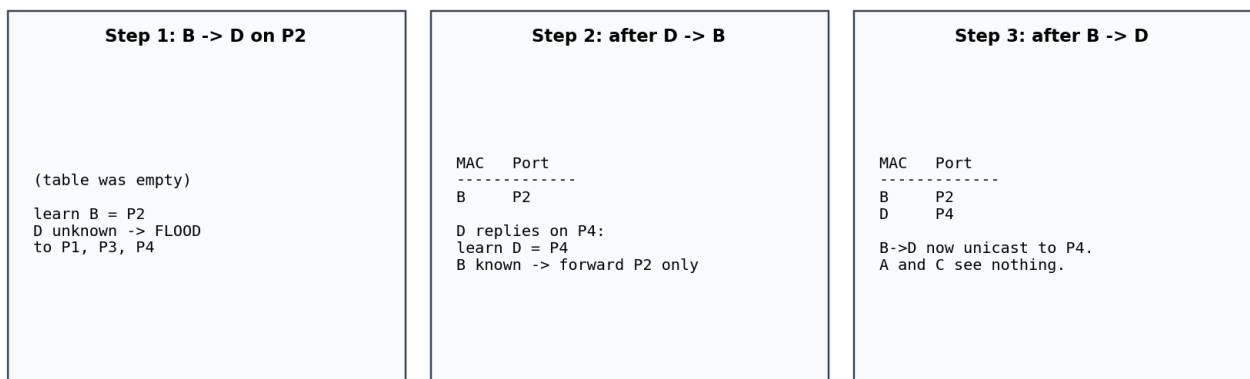


Fig. 3 Forwarding table construction in three steps

Event	Frame	Learning	Action
1	B -> D on P2	Add B = P2	D unknown: flood P1, P3, P4
2	D -> B on P4	Add D = P4	B known: forward P2 only
3	B -> D on P2	Refresh B	D known: forward P4 only
4	A -> FF:FF.. on P1	Add A = P1	Broadcast: flood P2, P3, P4

Cisco example for real hardware or Packet Tracer: `show mac address-table dynamic` lists the learned entries; `clear mac address-table dynamic` empties the table.

Q5. Hub, Switch and Bridge

Feature	Hub	Bridge	Switch
OSI layer	1 (Physical)	2 (Data link)	2 (some models also 3)
Forwarding	Regenerates bits and repeats them on every port	Forwards frames using a learned MAC table (software based)	Forwards frames using a MAC table in hardware (ASIC), cut-through or store-and-forward
Collision domains	One for the entire hub	One per port/segment (typically 2-4 ports)	One per port (none when full duplex)
Broadcast domains	One	One	One (per VLAN)
Duplex	Half only	Half	Half or full
Ports / speed	4-24, bandwidth shared	2-16, slower software path	8-48+, dedicated bandwidth per port
Practical use	Obsolete; occasionally used for debugging taps	Linking LAN segments of different media (e.g. wired to wireless); now built into APs	Standard LAN access/distribution device; VLANs, PoE, port security

Q6. CSMA/CD and Its Reduced Importance

Carrier Sense Multiple Access with Collision Detection is the MAC method of classic half-duplex (shared-medium) Ethernet. A station first listens; when the medium is idle it transmits and keeps listening. On detecting a collision it stops, emits a 32-bit jam signal, and tries again after a random delay chosen by **binary exponential backoff** (up to 16 attempts, after which it aborts).

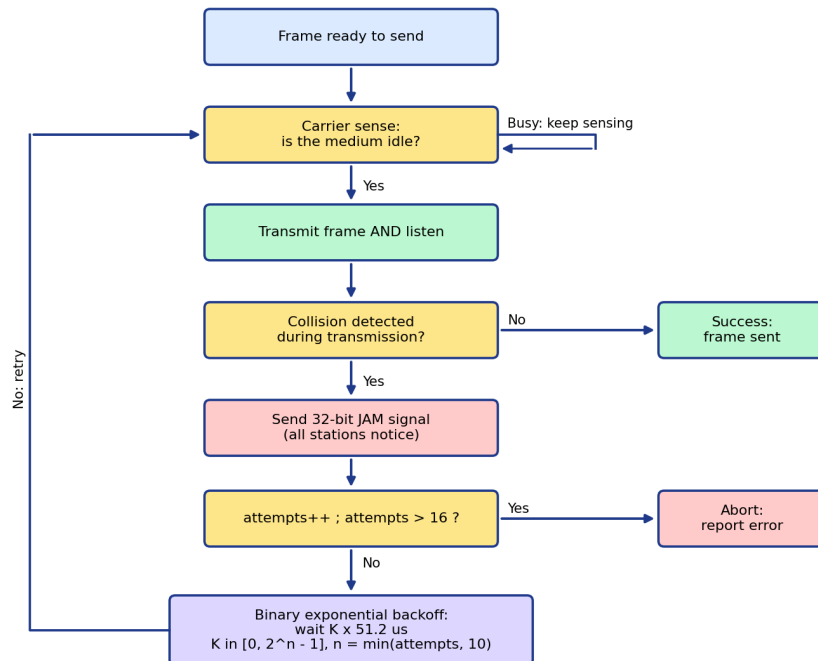


Fig. 4 CSMA/CD operation

Timing constraint: a sender must still be transmitting when the collision signal gets back to it, so the minimum frame is 64 B = 512 bits, i.e. a 51.2 us slot time at 10 Mb/s. This bounds the network diameter (about 2500 m at 10 Mb/s).

Why it matters less today

- **Switches** give every host a dedicated link, so each port is its own collision domain.
- **Full-duplex** operation uses separate transmit and receive paths, so sending and receiving happen together, collisions cannot occur and CSMA/CD is switched off.
- From 1 Gb/s upward (10GBASE-T and faster) the standards target full-duplex point-to-point links; half-duplex gigabit is defined but practically never deployed.
- Residual relevance: legacy hubs, and the unchanged MAC frame format and 64 B minimum, kept for backward compatibility. Congestion is now managed by switch buffering and flow control.

Q7. CSMA/CA in IEEE 802.11 and Comparison with CSMA/CD

A radio cannot reliably transmit and listen on the same channel at once, and the **hidden-node problem** means a station may never hear a competing sender. 802.11 therefore uses **collision avoidance** in its Distributed Coordination Function (DCF):

- Sense the channel using physical carrier sense plus virtual carrier sense, the **NAV**, which is set from the Duration field of overheard frames.
- If the channel stays idle for **DIFS**, pick a random **backoff** value from the contention window CW and count it down, pausing whenever the channel becomes busy.
- When the counter reaches zero, transmit. The receiver waits **SIFS** (shorter than DIFS, so ACKs win the channel) and returns an **ACK**.
- A missing ACK signals a collision or loss: CW is doubled (binary exponential) and the frame is retransmitted until the retry limit is reached.
- Optionally, **RTS/CTS** reserves the medium ahead of long frames, which counters hidden nodes and keeps the NAV consistent.

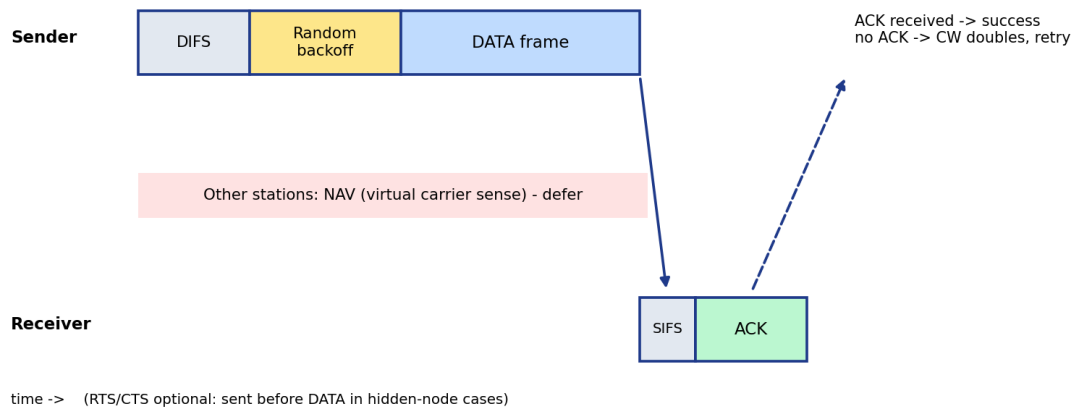


Fig. 5 CSMA/CA timeline with ACK and NAV

Aspect	CSMA/CD (802.3)	CSMA/CA (802.11)
Objective	Detect a collision as <i>it happens</i> and recover fast	Avoid collisions in advance, because they cannot be detected
Medium	Wired, half-duplex shared cable	Shared radio channel
Collision handling	Abort at once, send jam, back off	Finish the frame, wait for the missing ACK, back off with doubled CW
Acknowledgement	None at layer 2	Mandatory link-layer ACK for unicast
Hidden node	Not an issue	Real issue; handled by RTS/CTS and NAV
Overhead	Low	High (DIFS, backoff, SIFS + ACK, optional RTS/CTS)
Modern status	Largely unused (full-duplex switches)	Still central to Wi-Fi (alongside OFDMA in 802.11ax)

Q8. Design of a Small WLAN for a College Laboratory

Scenario. One lab of about 18 m x 9 m seating 24 students plus an instructor, with a wired uplink in the server rack. Requirements: dependable access for laptops and phones, coverage everywhere, and isolation from the campus admin network.

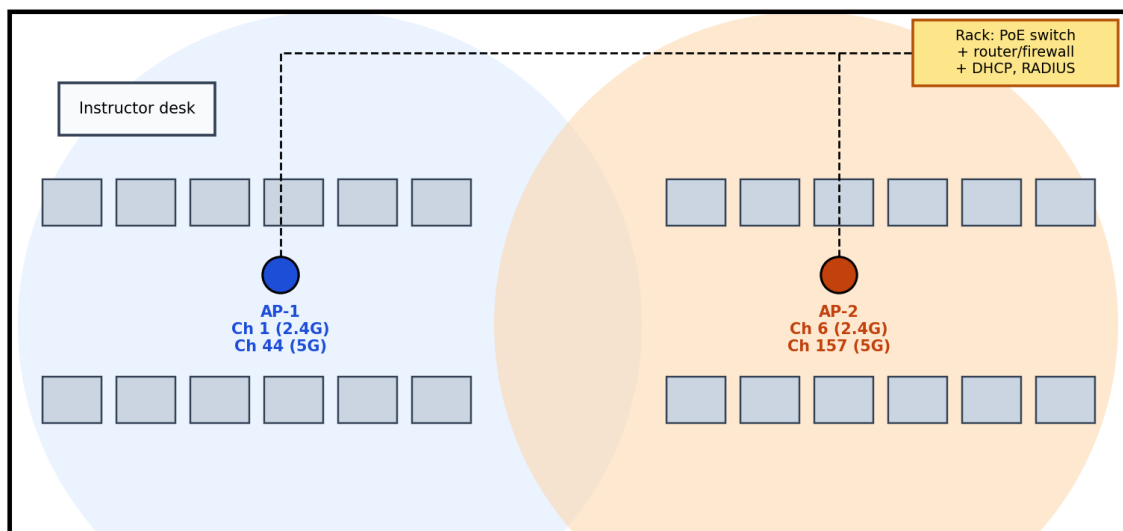


Fig. 6 Lab layout (about 18 m x 9 m, 24 seats): ceiling APs ~ 8 m apart; dashed = Cat6 PoE uplinks; circles = approx. cell

Fig. 6 Floor plan with access point placement

Design item	Specification
AP count and placement	2 dual-band Wi-Fi 6 (802.11ax) APs, ceiling-mounted at ~2.7 m, one over each half of the room (~8 m apart); ~12-15 clients per AP. Keep clear of metal cabinets and the rack. Powered by PoE (802.3af/at) from a managed switch over Cat6.

Design item	Specification
SSIDs	CS-LAB (students/staff, WPA3-Personal or WPA2/WPA3-Enterprise with 802.1X); CS-LAB-Guest (optional, client isolation, bandwidth cap). A hidden SSID is not used because it adds no real security.
Channels	2.4 GHz: non-overlapping channels 1 and 6 (20 MHz) - AP-1 on 1, AP-2 on 6. 5 GHz: AP-1 on 44 , AP-2 on 157 , 40 or 80 MHz width. Steer clients to 5 GHz (band steering) and keep 2.4 GHz transmit power low.
IP addressing	VLAN 30, 10.30.30.0/24 ; gateway 10.30.30.1; APs static 10.30.30.2-.3; DHCP pool .50-.150 (101 leases for 24-50 devices), 8-hour lease; DNS from the campus resolver. Guest VLAN 40 on 10.30.40.0/24 with NAT.
Security	WPA3 (or WPA2-AES, never WEP/TKIP); disable WPS; change default admin passwords; HTTPS/SSH management on a separate management VLAN 99; regular firmware updates; rogue-AP detection; firewall rules between lab and admin VLANs; logging via RADIUS. MAC filtering is optional, not a substitute.
Capacity and QoS	Each AP delivers ~500 Mb/s real throughput on 5 GHz; enable QoS (WMM) for voice/video if used. Run a site survey with a tool such as Wi-Fi Analyzer or Ekahau to confirm RSSI >= -65 dBm everywhere.

Q9. Mobile IP and the Need for Mobility Support

Problem. An IP address plays two roles - identity and location. Routers forward on the network prefix, so a host that moves to another network and adopts a new address breaks its TCP connections and can no longer be reached at its old address. Mobile IP (RFC 5944 for IPv4, RFC 6275 for IPv6) lets a node keep a single permanent **home address** while roaming, invisibly to applications.

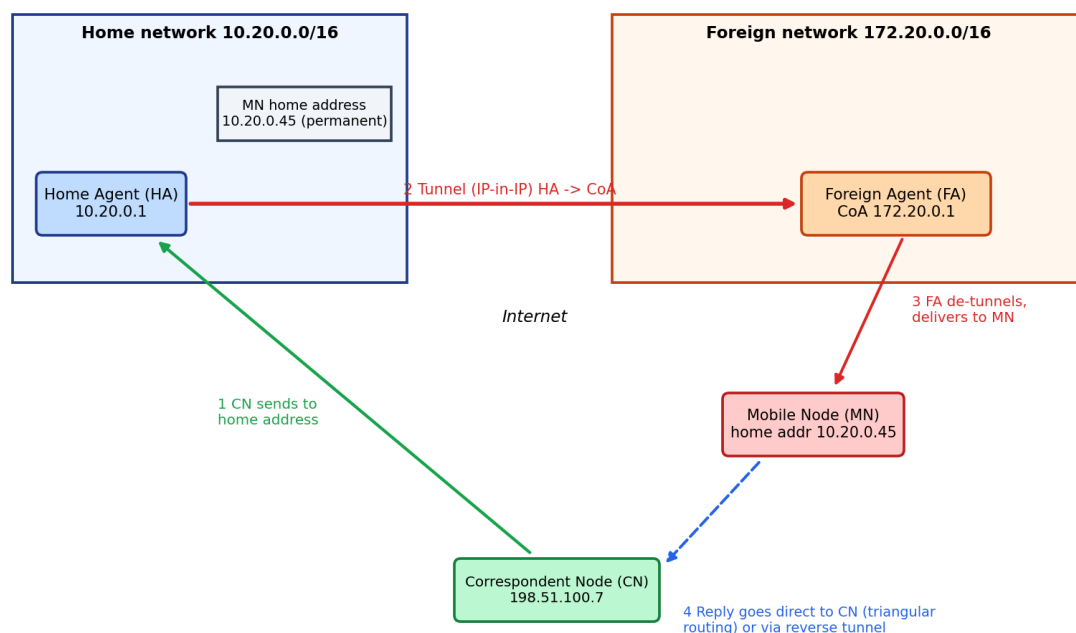


Fig. 7 Mobile IP architecture and packet flow

Term	Meaning
Mobile Node (MN)	The roaming host that owns a permanent home address.
Home Agent (HA)	Router on the home network that intercepts packets addressed to the MN and tunnels them to its current location.
Foreign Agent (FA)	Router on the visited network that supplies a care-of address and hands over de-tunnelled packets.
Care-of Address (CoA)	Temporary address marking the MN's present location (FA CoA or co-located CoA).

Operation: (1) **Agent discovery** - HA/FA broadcast advertisements (ICMP router discovery) and the MN realises it is away from home. (2) **Registration** - the MN registers its CoA with the HA, via the FA (UDP port 434). (3) **Tunnelling** - the HA encapsulates packets for the home address in IP-in-IP towards the CoA; the FA/MN decapsulates them. (4) The MN answers directly (triangular routing) or through a reverse tunnel when ingress filtering is applied.

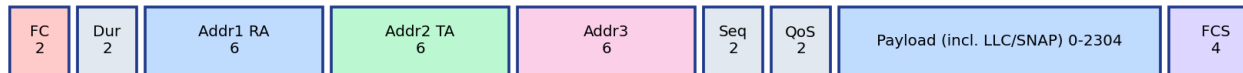
Issues: triangular routing adds delay (route optimisation in MIPv6 avoids it); registration must be authenticated to prevent hijacking; tunnelling adds overhead. **Need:** laptops, phones, vehicles and IoT devices move between access networks yet expect calls, downloads and VPNs to survive. Inside one Wi-Fi ESS, roaming between APs is handled at Layer 2 (802.11r/k/v) without Mobile IP; Mobile IP is for moves across *IP subnets*.

Q10. Packet-Level Comparison: Wired Ethernet vs Wireless LAN Frame

IEEE 802.3 Ethernet II frame (14 B header)



IEEE 802.11 QoS data frame (26 B MAC header, up to 32 with Addr4)



Addr4 appears only in WDS frames (ToDS=1, FromDS=1). Control frames: ACK = 14 B (FC, Dur, RA, FCS); RTS = 20 B; CTS = 14 B.

Fig. 9 802.3 vs 802.11 data frame layout

Feature	Ethernet II (802.3)	IEEE 802.11 data frame
Header size	14 B (6+6+2)	24 B (26 with QoS; 30/32 with Addr4)
Address fields	2: destination, source	Up to 4; meaning depends on To-DS/From-DS bits
Control info	None (type only)	Frame Control: version, type/subtype, To-DS, From-DS, retry, power management, protected, etc.
Duration/NAV	No	Yes, 2 B, used for virtual carrier sense
Sequence control	No	Yes - sequence and fragment numbers for retransmission and fragmentation
Max payload	1500 B	2304 B (MSDU)
Protocol identification	EtherType field	LLC/SNAP header in the body
Acknowledgement	None	Mandatory 14 B ACK frame for unicast
Frame types	One basic type	Management (beacon, probe, auth, assoc), control (RTS, CTS, ACK), data
Encryption	None at L2 (MACsec optional)	Built in: CCMP/GCMP (WPA2/WPA3) - Protected bit set
Capture	Any NIC	Needs monitor mode (e.g. Linux <code>airmon-ng</code>) to expose 802.11 headers; otherwise Wireshark shows synthetic Ethernet frames

Address fields in 802.11 (infrastructure mode)

Direction	To-DS	From-DS	Addr1 (RA)	Addr2 (TA)	Addr3
Station -> AP	1	0	BSSID (AP)	Source (SA)	Destination (DA)
AP -> Station	0	1	Destination (DA)	BSSID (AP)	Source (SA)

Documented example. A laptop (MAC 5c:ba:37:12:ab:01) pings a wired server through the AP with BSSID 6c:4b:90:33:55:77. The wireless-side frame (Wireshark filter `wlan.fc.type_subtype == 0x28`, QoS Data) has To-DS=1, Addr1=6c:4b:90:33:55:77, Addr2=5c:ba:37:12:ab:01, Addr3=server MAC, and is acknowledged by a 14 B ACK from the AP. Once the AP bridges it onto the wire, the same payload appears as an Ethernet II frame with Destination=server MAC (e4:5f:01:aa:bc:09), Source=5c:ba:37:12:ab:01 and EtherType 0x0800. The AP therefore performs the 802.11-to-802.3 frame translation. These values are a documented sample; capture your own with the filters above and attach the screenshots.

Comparison Summary

Topic	Wired (Ethernet)	Wireless (Wi-Fi)
Access method	CSMA/CD (half duplex) / none (full duplex)	CSMA/CA with ACK
Reliability	Link layer drops bad frames	Link layer retransmits
Addressing	2 MAC addresses	3-4 MAC addresses
Security	Relies on physical access	Encryption and authentication essential

Repository Structure (submission)

```
assignment3-repo/  
|-- README.md  
|-- report/Assignment3_Report.pdf  
|-- diagrams/ (Figures 1-7, 9 as PNG)  
|-- screenshots/ (your Wireshark screenshots)  
|-- captures/ (ethernet.pcapng, wlan.pcapng)  
`-- design/ (WLAN design notes)
```

References

- IEEE Std 802.3-2022, IEEE Standard for Ethernet.
- IEEE Std 802.11-2020, Wireless LAN MAC and PHY Specifications.
- IEEE Std 802.2, Logical Link Control.
- C. Perkins (ed.), RFC 5944, IP Mobility Support for IPv4, Revised, IETF, 2010; D. Johnson et al., RFC 6275, Mobility Support in IPv6, 2011.
- B. A. Forouzan, Data Communications and Networking, McGraw-Hill.
- J. Kurose and K. Ross, Computer Networking: A Top-Down Approach, Pearson.
- A. S. Tanenbaum and D. Wetherall, Computer Networks, Pearson.
- Wireshark User's Guide, <https://www.wireshark.org/docs/>